# IAS Physics

## Topic 1: Mechanics

### 1A: Motion (1. Velocity and Acceleration)

1) For uniformly accelerated motion in one direction, the following five **SUVAT** equations apply:

displacement (**s**), initial velocity (**u**), final velocity (**v**), acceleration (**a**), time (**t**)

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$$s = \frac{u + v}{2} \times t$$

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$$v = u + at$$

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$$s = ut + \tfrac{1}{2} at^2$$

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$$s = vt - \tfrac{1}{2} at^2$$

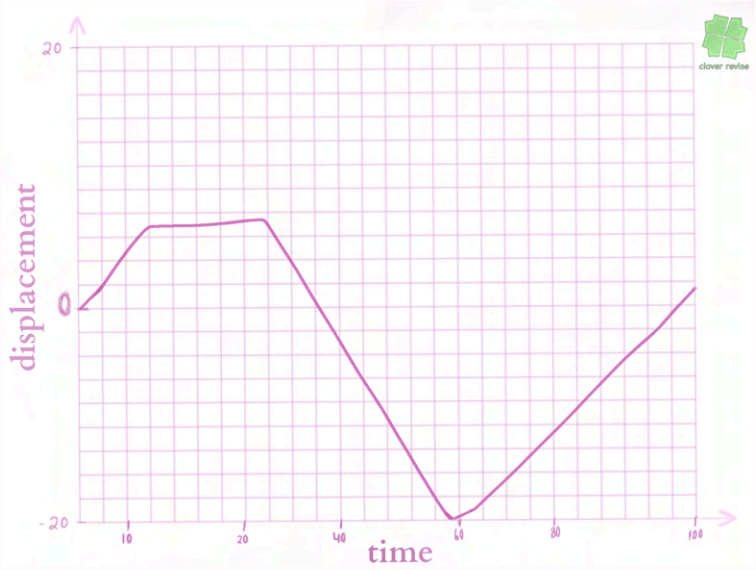
$$v^2 = u^2 + 2as$$

## 1A: Motion (2. Motion Graphs)

2 + 3)

a) For a DISPLACEMENT-TIME graph, the gradient is the **velocity**.

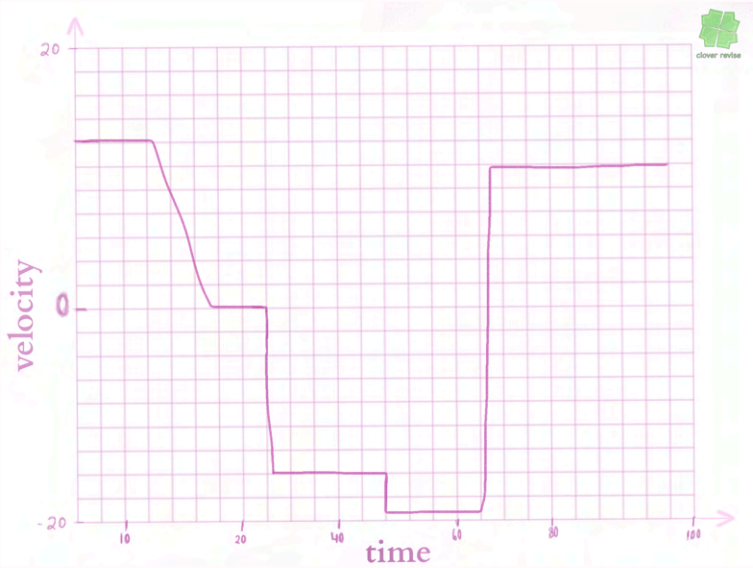
gradient = **distance (m) ÷ time(s) = velocity (m/s)**




a) For a VELOCITY-TIME GRAPH:

- the gradient is the **acceleration**.

gradient = **final speed (m/s) - initial speed (m/s)** / **time (s)** = **acceleration (m/s²)**




- the area under the graph is the **distance travelled.**

area = **velocity (m/s) x time(s) = distance (m)**

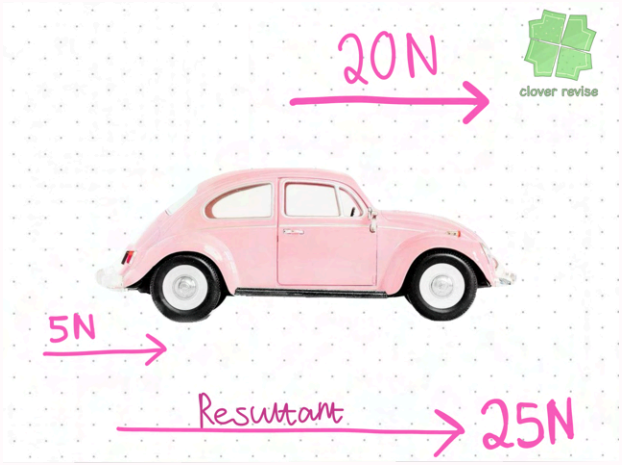
1A: Motion (3. Adding Forces)

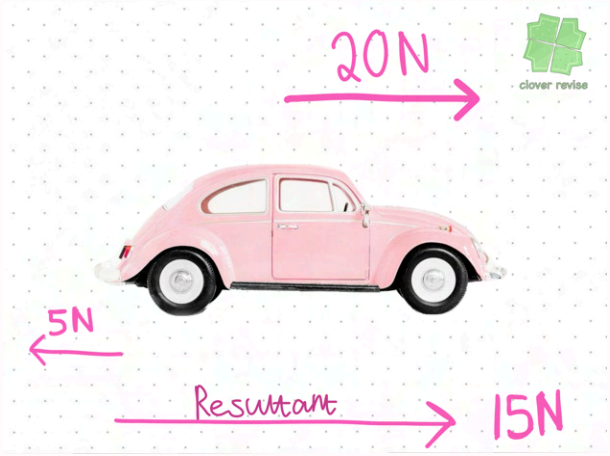
4) A **vector** quantity, such as a force, has both magnitude and direction.

| SCALAR | VECTOR |
|---|---|
| distance | displacement |
| speed | velocity |
| mass | weight |
| energy | acceleration |
| density | force |
| power | impulse |
| volume / area / length | momentum |

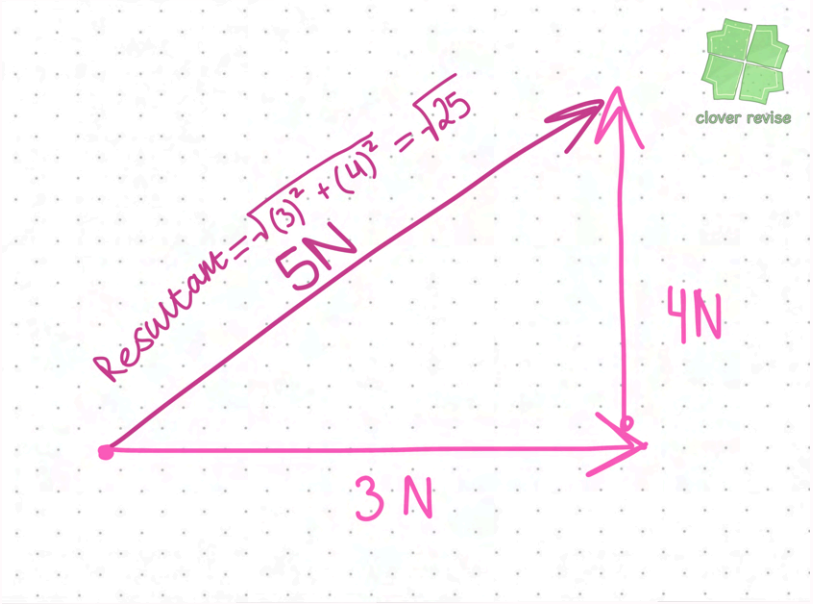
| work done | friction / drag force |
|---|---|

Resultant force is the **vector** sum of all forces acting on an object.

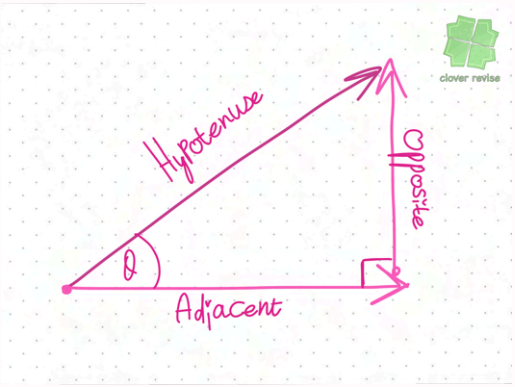


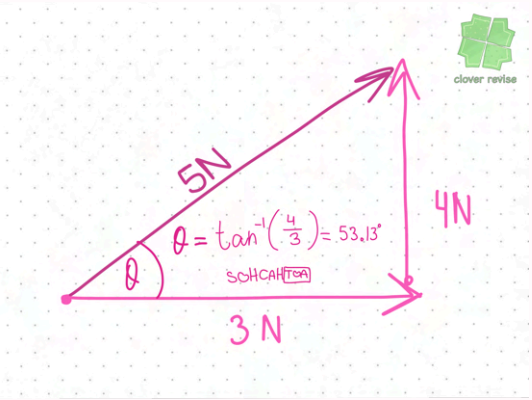




5) The magnitude of the resultant force of two perpendicular forces, at <u>right angles</u> to one another, is calculated using **Pythagoras' theorem**.



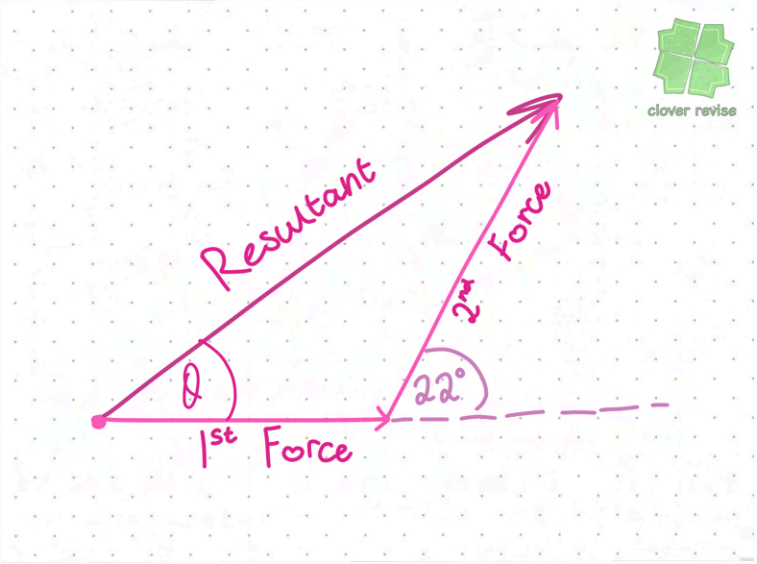

**Trigonometry** (**SOHCAHTOA**) must be used to find the **direction** of the force.



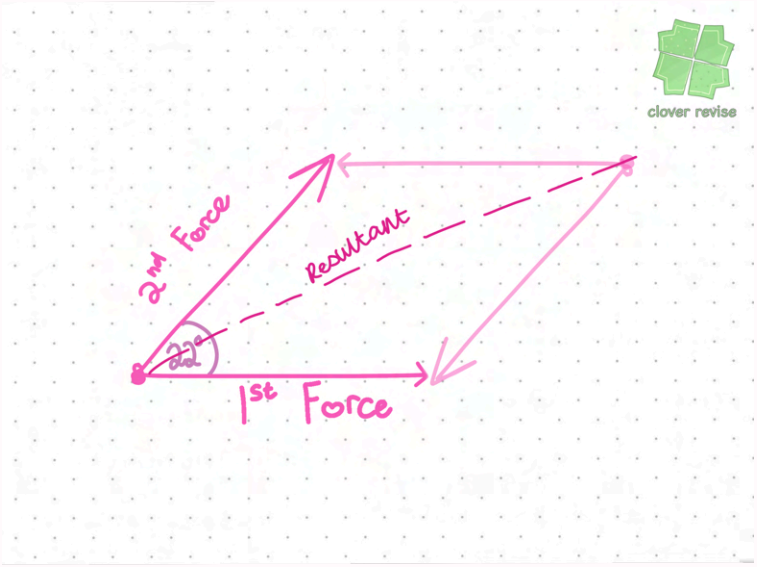




sinθ (**s**), opposite (**o**), hypotenuse (**h**), cosθ (**c**), adjacent (**a**), tanθ (**t**)

6) For two **non-perpendicular** (planar) forces, a scale-diagram must be drawn. Draw one force after the other, and the resultant will be from the start of the first force until the end of the second one. It should make a triangle.

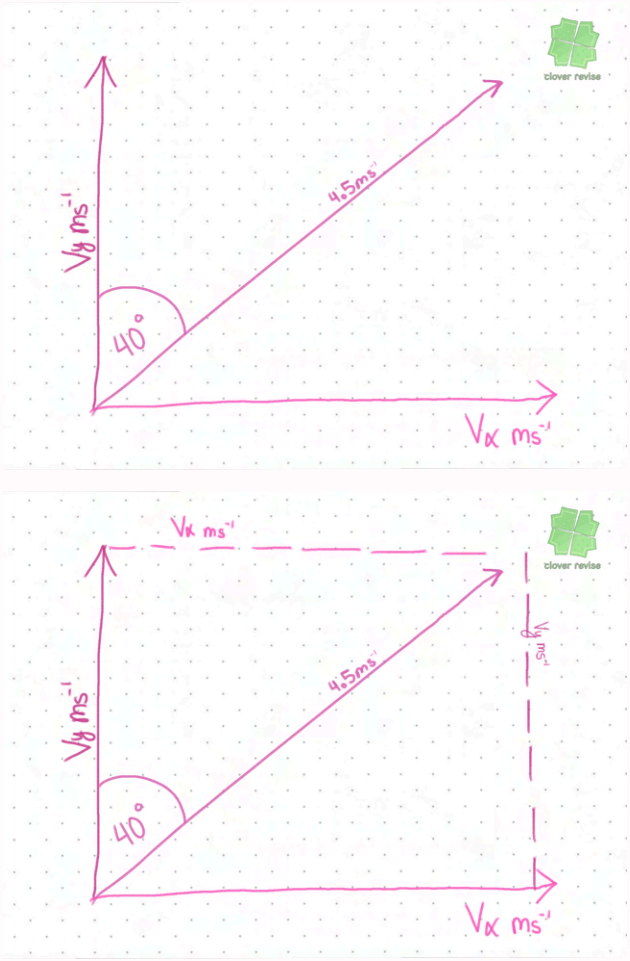



Alternatively, the **parallelogram rule** can be used. Draw both forces to scale, starting from the same point and at the correct angle, using their lengths to represent their magnitudes. Then draw the same forces parallel to the first two, so that a parallelogram is made. The resultant force will be the diagonal across the parallelogram from its starting point.

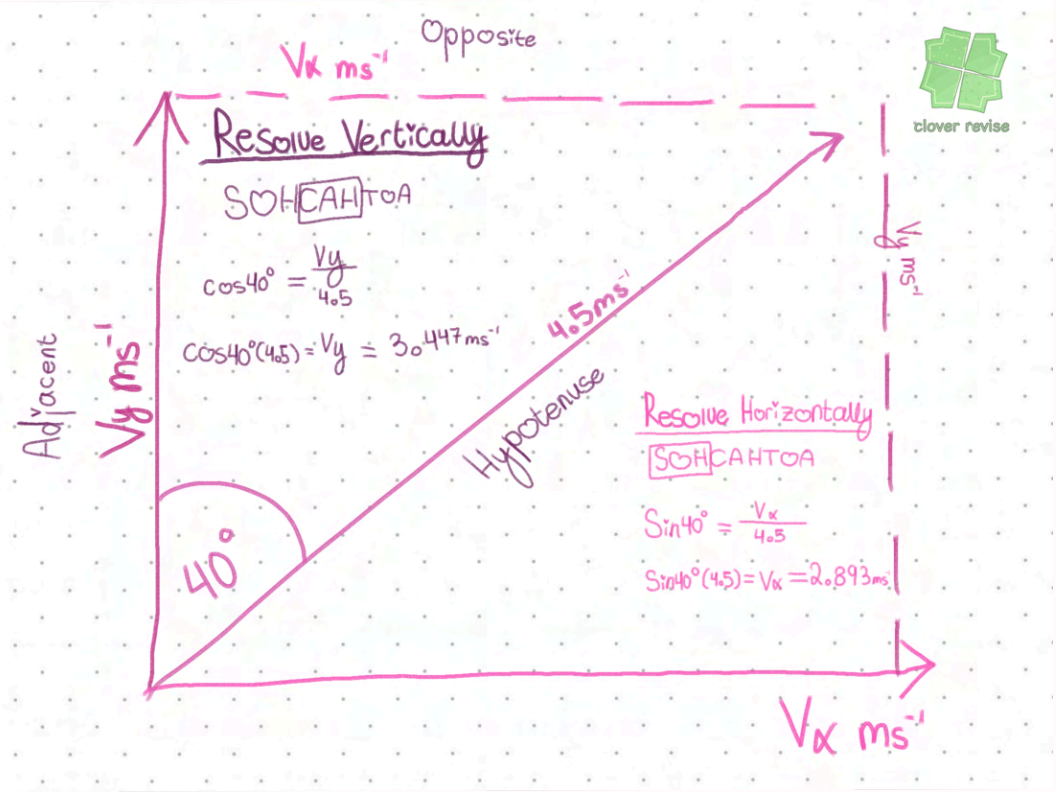


1A: Motion (7. Resolving Vectors)

7) A vector can be split into vertical and horizontal components that are at right angles to one another. This is called **resolving vectors**, and the resolved vectors start at the same point as the original vector.




**Vy** is the vertical component, and **Vx** is the horizontal component.
We can apply the **parallelogram rule** when resolving vectors.




We resolve vectors using trigonometry, as shown above.

1A: Motion (8. Projectiles)

When an object is thrown through the air, it follows a **projectile** motion.

**Horizontal Throws**

An object thrown horizontally, due to gravity, will curve downwards in a parabola shape, as shown below.




- Objects thrown **horizontally** have an initial vertical velocity of 0.

**H**( u = 0 )

- There is no acceleration **horizontally**.

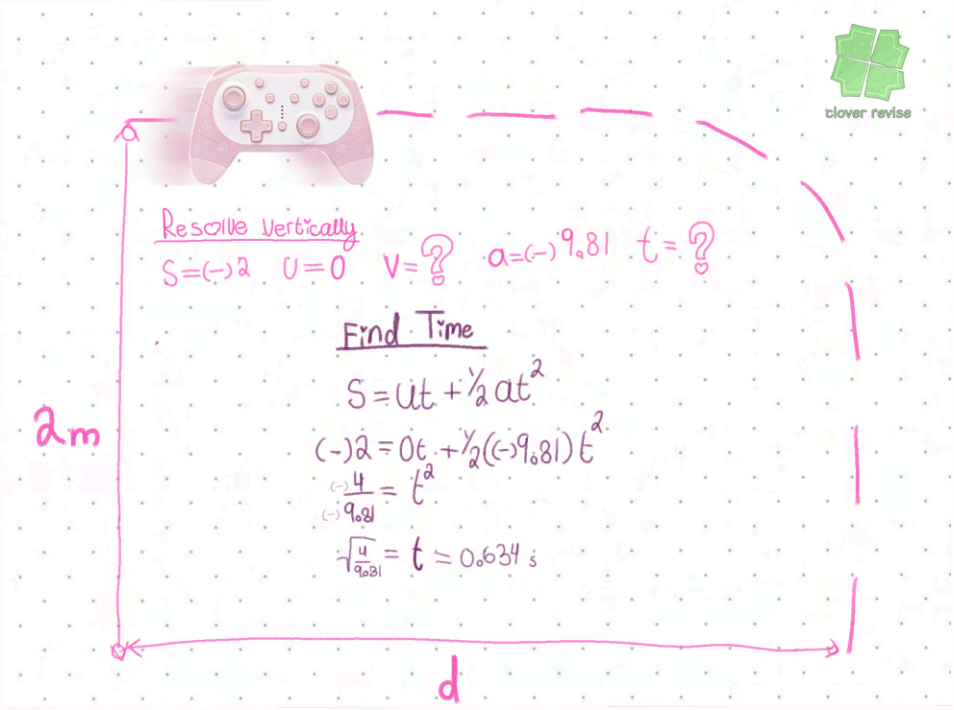
**H**( a = 0 )

- **Vertically**, acceleration is either positive or negative 9.81, depending on which direction you consider as positive.

**V**( a = (-)9.81 )

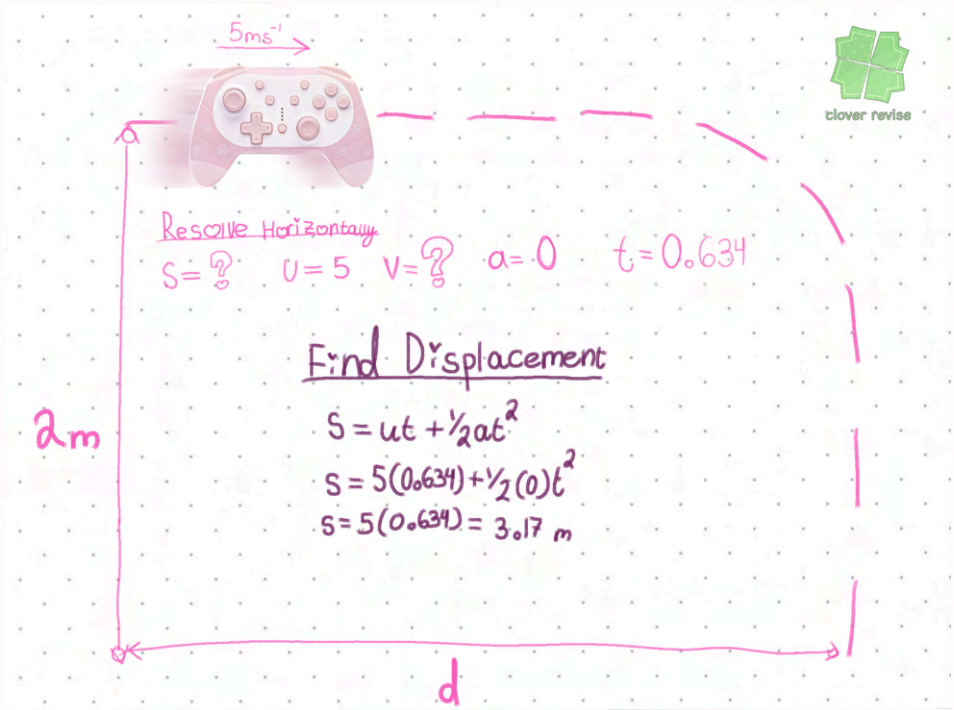
We must use **SUVAT** equations and **resolving** to solve these problems.

For example, if we wanted to find the horizontal distance travelled by the controller, we would first need to resolve vertically to find the time of the projection. If not, there would not be enough information to use **SUVAT** horizontally.




**Note**: Remember to choose a direction as positive or negative.
As the object is falling, downwards acceleration due to gravity and displacement should either both be positive or negative.

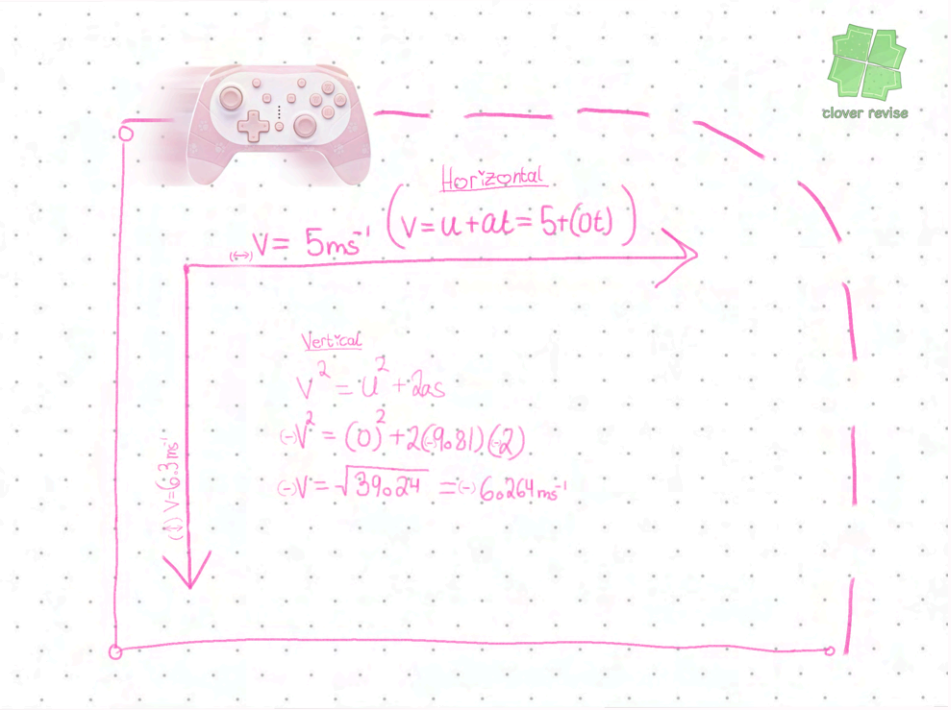
We can then resolve <u>horizontally</u> to find the controller's <u>displacement</u>.



To find the controller's speed on landing, we would have to calculate the magnitude of the resultant vector by combining the vertical and horizontal components for final velocity.

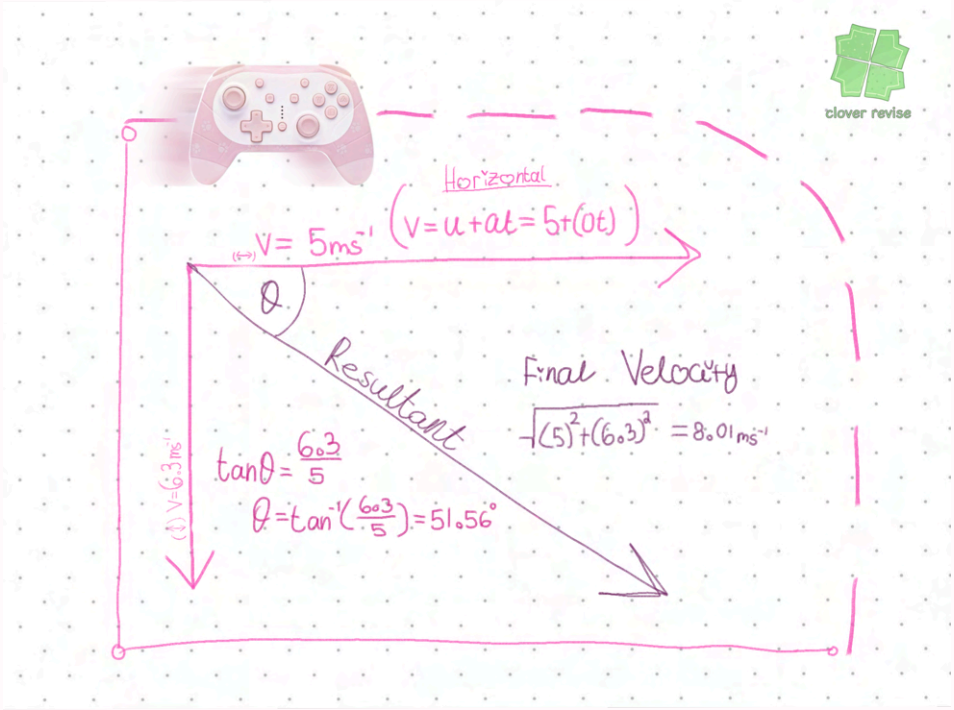
As we know that there is no horizontal acceleration, the final velocity will be equal to the initial velocity. In this case, that is 5m/s.

However, **SUVAT** must be used to find the final velocity vertically.




When we have both components, we can then recombine them into their resultant vector by using **Pythagoras' Theorem**.

Additionally, **trigonometry** can be used to find the angle at which the object
is flying when it lands.




**<u>Vertical Throws</u>**

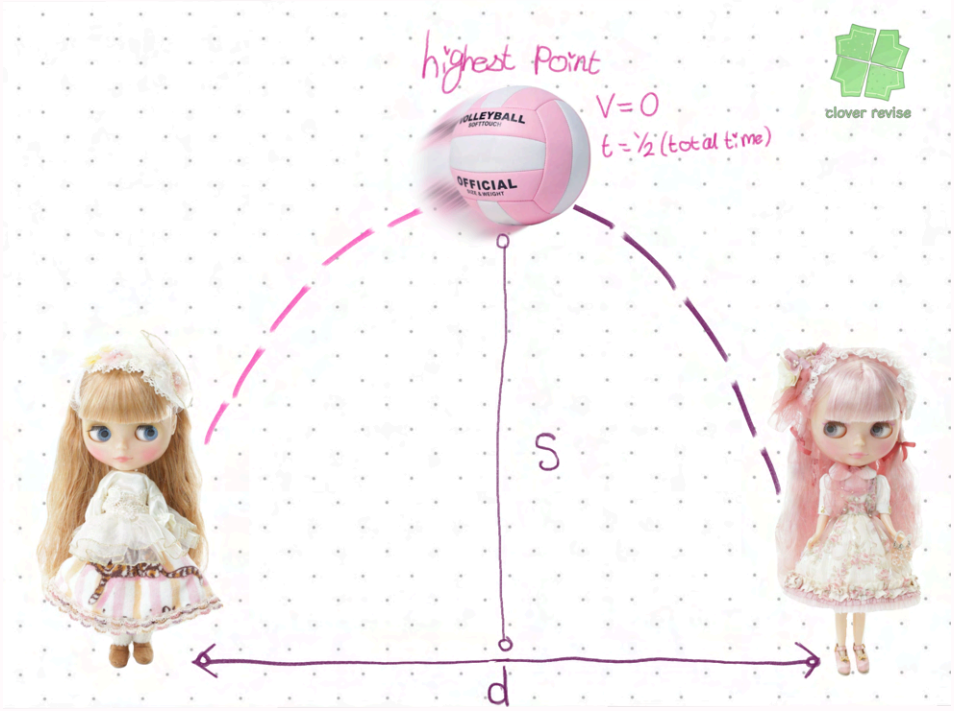
An object thrown vertically is assumed to have a symmetrical trajectory.

- The <u>velocity at the highest point is o</u>.

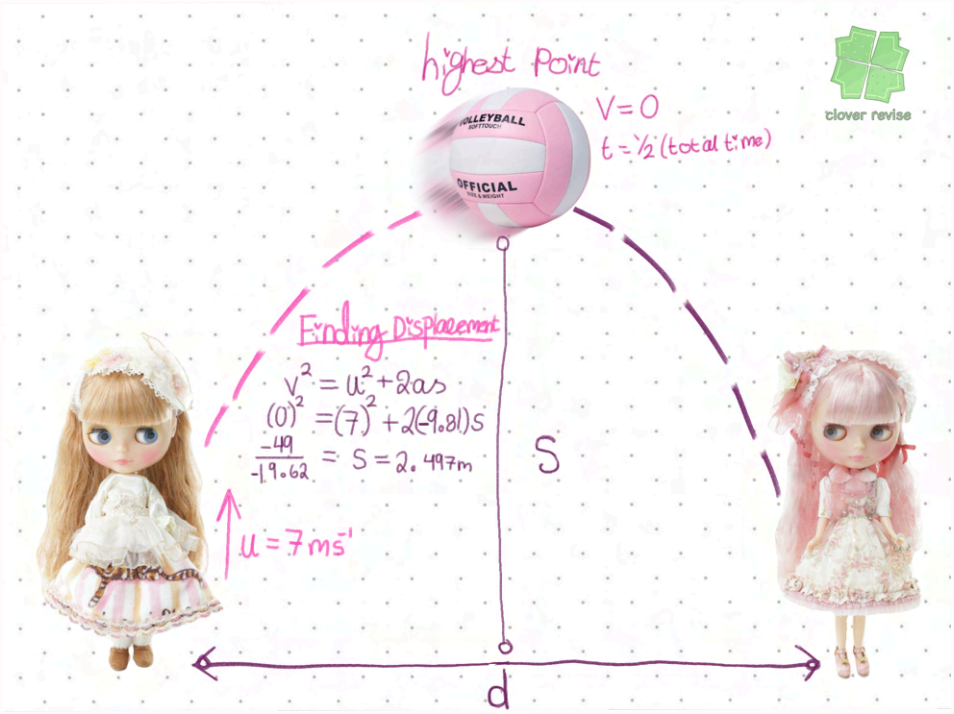
  ( v = o )

- The time taken to get the <u>highest point is half the time</u> of the entire flight.

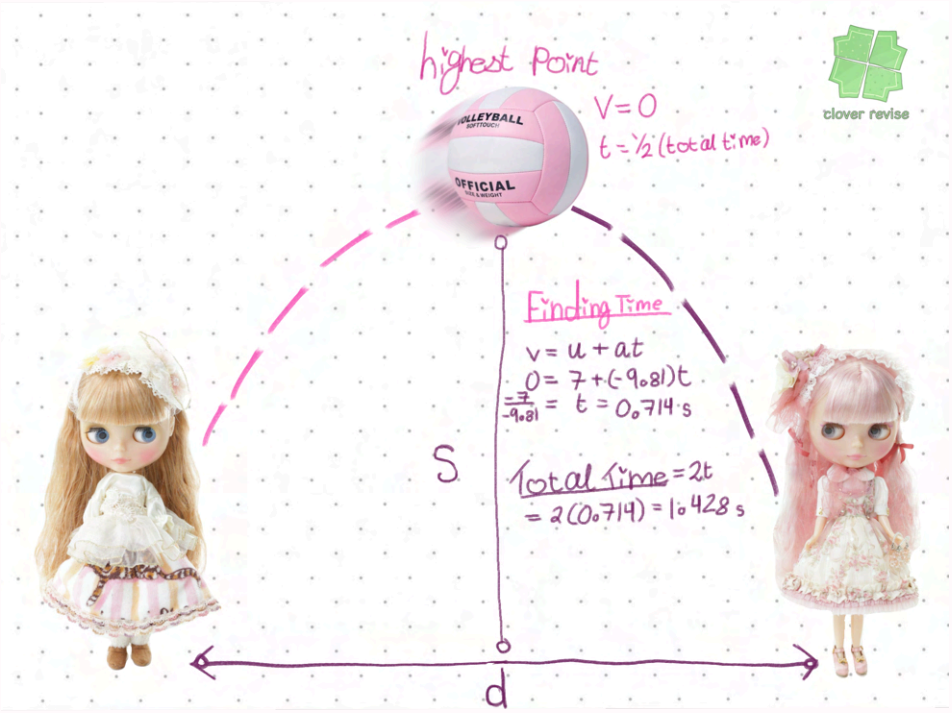
  ( t = ½ total time )




We can use **SUVAT** to find both the ball's <u>height</u> at its highest point, and the <u>distance</u> between both friends.

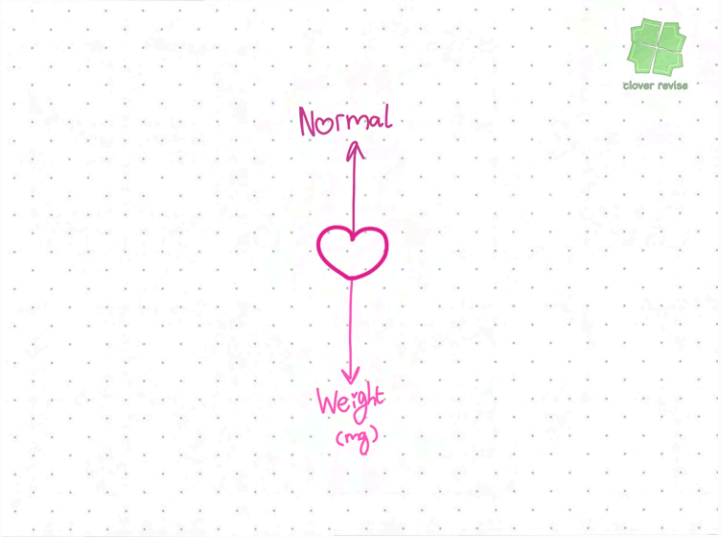


**Note**: As the ball is initially thrown upwards, acceleration due to gravity is in the opposing direction.




8) A free-body force diagram can be drawn to show all of the forces acting on an object and where those forces act.

For an object in equilibrium, the arrow lengths of opposing forces (e.g. weight and normal reaction force) are equal.




1A: Motion (5. Newton's Laws of Motion)

9 + 10 + 12) **N1**: A stationary object needs a resultant force to be moved.

An object will continue to move at the same speed in the same direction, unless a resultant force acts on it.

If there is no resultant force, an object's motion remains unchanged.

**N2**: Resultant Force (**F**) = Mass (**m**) x Acceleration (**a**)

**F** (N) = **m** (kg) x **a** (m/s²)

For the force accelerating a mass downwards (weight),

Weight (**W**) = Mass (**m**) x Gravity (**g**)

W = mg **OR** F = mg

g = W ÷ m **OR** g = F ÷ m

N3: For every action, there is an <u>equal and opposite</u> reaction.

When object A causes a force on object B, object B causes an equal force in the opposite direction to act on object A.

11) **Core Practical 1:** Determining the Acceleration of a Freely-Falling Object




Method A) Sphere Falling through a Trap Door

1. Drop the sphere from rest and record the time taken (**t**) for it to fall through the trap door.
2. Repeat two more times and find the mean value for **t**.
3. Measure and record the height fallen by the sphere (**h**).

5. Repeat **steps 1-3** at at least six varying heights.

6. Half of the range in the readings for **t** are the uncertainty in **t**.

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Method B) Dowel Passing through a Light Gate

1. Measure the length of the dowel.

2. Drop the dowel from rest and record the time taken (**t**) for it to pass through the light gate.

3. Repeat two more times and find the mean value for **t**.

4. Measure and record the height fallen by the object (**h**).

5. Repeat **steps 1-3** at at least six varying heights.

6. Half of the range in the readings for **t** are the uncertainty in **t**.

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**SAFETY:**

- Clamp the stamp securely.
- Keep hands and face away from falling objects.
- Turn off electromagnet between drops to avoid overheating and burns.

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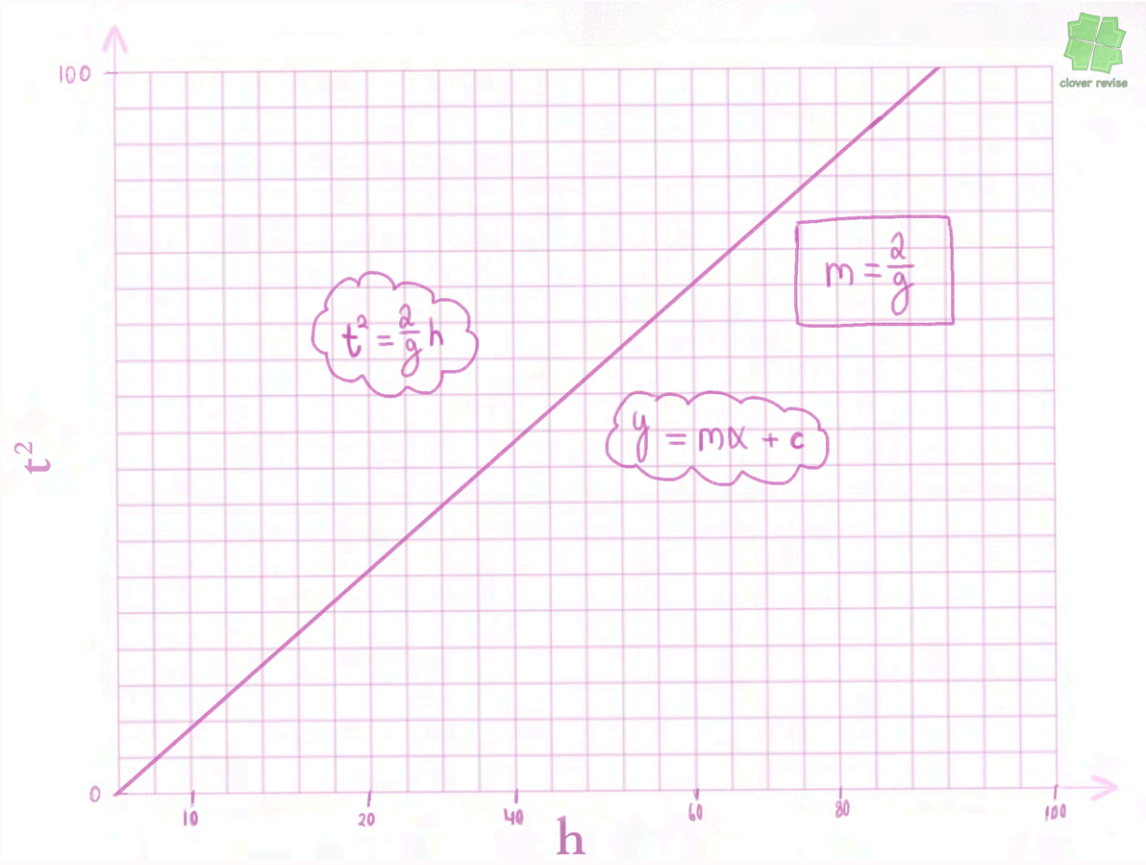
For method A,

The **SUVAT** equation ( $s = ut + \frac{1}{2} at^2$ ) should be used, wherein $u = 0$

(object dropped from rest), $s = h$ (height fallen), and $a = g$ (object is accelerating downwards).

This equation can then be rearranged as ( $t^2 = 2h / g$ ) and when we plot a graph of **$t^2$** (**y**-axis) against **h** (**x**-axis), we will get a straight line with the gradient $2/g$.

$y = mx + c$ **THEREFORE** $t^2 = 2/g \times h$



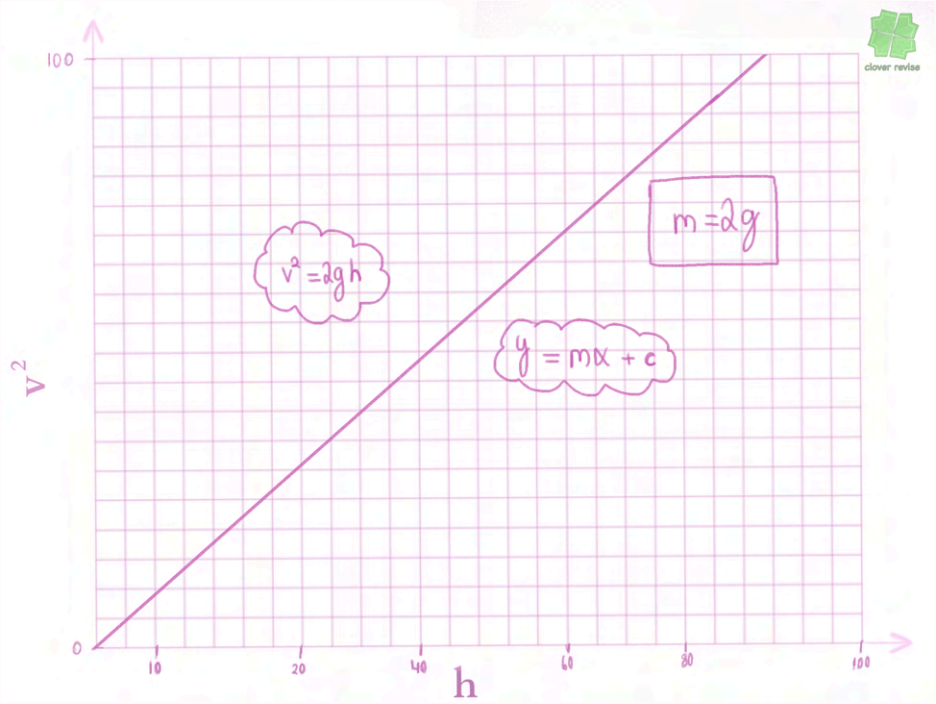

For method B,

The **SUVAT** equation ( $v^2 = u^2 + 2as$ ) should be used, wherein $u = 0$, $s = h$, and $a = g$.

This equation can then be rearranged as ( $v^2 = 2gh$ ) and when we

plot a graph of $v^2$ ($y$-axis) against **h** ($x$-axis), we will get a straight line with the gradient 2**g**.

$$y = mx + c \textbf{ THEREFORE } v^2 = 2g \times h$$




## 1C: Momentum

13) **Momentum** is a measure of an object's motion.

Momentum (**p**) = Mass (**m**) x Velocity (**v**)

**p** (kgm/s) = **m** (kg) x **v** (m/s)

Change in momentum divided by time is therefore mass x acceleration ( **v** / **t** = **a** ). This means resultant force is equal to change in momentum divided by time.

$$\Delta p \div \Delta t = ma = F$$

## 1C: Momentum (2. Conservation of Linear Momentum)

14) The principle of conservation of linear momentum states that the sum total of momenta for objects before a collision is equal to the sum total after the collision.

Remember that direction must be accounted for.

$$\Sigma \text{ Momentum Before } (\mathbf{p} \text{ before}) = \Sigma \text{ Momentum After } (\mathbf{p} \text{ after})$$

If a stationary object explodes, the total momentum after must be zero, because a stationary object will have a velocity and, hence, momentum of zero.

## 1A: Motion (4. Moments)

15) The **moment** of a force is the tendency to cause rotation.

$$\text{moment} = \text{force } (\mathbf{F}) \times \text{perpendicular distance from pivot } (\boldsymbol{x})$$

$$\textbf{moment } (\text{Nm}) = \mathbf{F} \ (\text{N}) \times \boldsymbol{x} \ (\text{m})$$

For an object in **equilibrium**:

- Resultant force is zero.
- Resultant rotational force is zero (clockwise moments = anti-clockwise moments).

An object in equilibrium will either remain stationary or continue moving at the same velocity.

The **centre of gravity** is the point at which an object's weight appears to act.
- For a symmetrical object, it lies on the point of intersection for all lines of symmetry.

## 1B: Energy (2. Work and Power)

17 + 21) **Work done** is a measure of the amount of energy transferred.

Work Done (**W**) = energy (**E**) = Power (**P**) x Time (**t**)
**W** (J) = **E** (J) = **P** (W) x **t** (s)

When a force is used to transfer energy mechanically, the amount of work done is calculated by:

work done (**W**) = Force (**F**) x Distance Moved in Force's Direction (**Δs**)
**W** (J) = **F** (N) x **Δs** (m)

### Work Done by Forces at an Angle

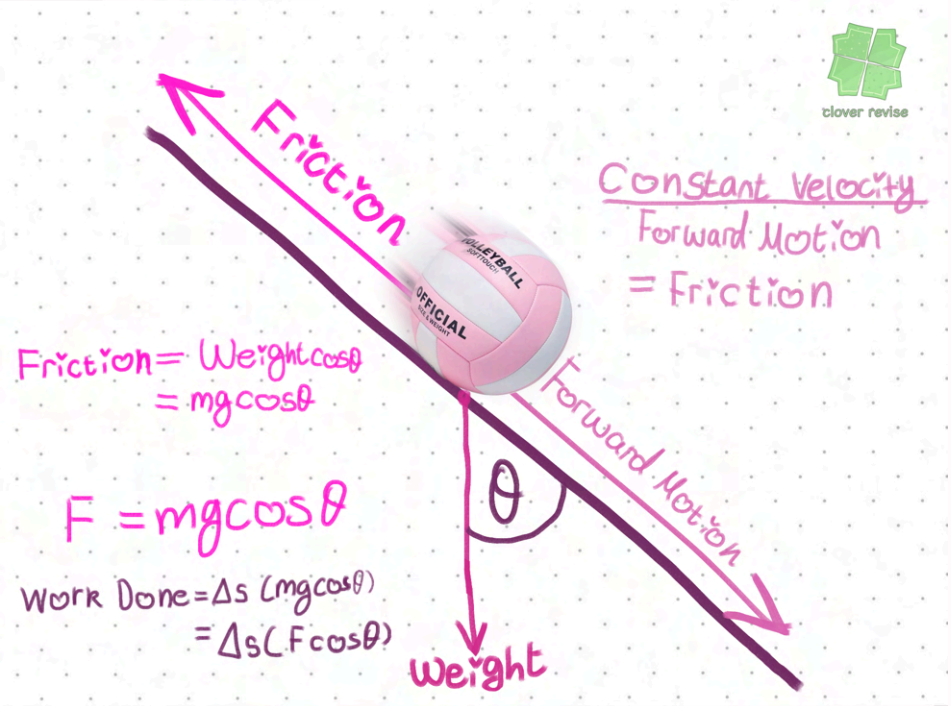
If an object is sliding down, then friction will act upwards (opposing the direction of motion), meaning work must be done against friction.

As there is no acceleration, the friction must have the same magnitude as the forward motion, so that there will be no resultant force (**N2**).

The forward motion acts at an angle of θ to the component of weight, and

is therefore equal to **mgcosθ**, which will also be the magnitude of friction.

Therefore, work done will be equal to Δs x **mgcosθ**.
The general formula for work done at an angle is then Δs x **Fcosθ**.




1B: Energy (1. Gravitational Potential and Kinetic Energies)

18) **Kinetic energy** is the energy an object gains due to its movement.

kinetic energy (**$E_k$**) = ½ x Mass (**m**) x Velocity² (**$v^2$**)

**$E_k$** (J) = ½ x **m** (kg) x **$v^2$** (m/s)

19) **Gravitational potential energy** is the energy an object gains due to its position in a gravitational field.

gravitational potential energy (**Egrav**) = Mass (**m**) x Gravity (**g**) x Change in Height (**Δh**)

$$\mathbf{E_{grav}}\ (\text{J}) = \mathbf{m}\ (\text{kg}) \times \mathbf{g}\ (\text{N/kg}) \times \Delta\mathbf{h}\ (\text{m})$$

Work done can also be found by multiplying energy and time.

20) **The principle of conservation of energy** states that energy can neither be gained nor lost. Total energy before and after the transfer must be equal.

An object that has been dropped will have its $\mathbf{E_{grav}}$ transferred to $\mathbf{E_k}$ as it falls, and an object being thrown up will have its $\mathbf{E_k}$ transferred to $\mathbf{E_{grav}}$.

Therefore: $\mathbf{E_k} = \frac{1}{2} \times \mathbf{m} \times \mathbf{v^2} = \mathbf{E_{grav}} = \mathbf{m} \times \mathbf{g} \times \Delta\mathbf{h}$

The **m**ass will cancel on either side, leaving us with:

$$\mathbf{E_k} = \frac{1}{2} \times \mathbf{v^2} = \mathbf{E_{grav}} = \mathbf{g} \times \Delta\mathbf{h}$$

If we now want to find out how high an object will rise if projected upwards, we can use the equation:

$$\frac{1}{2} \times \mathbf{v^2} \div \mathbf{g} = \Delta\mathbf{h}$$

Or, if we wanted to find out how fast an object will drop from a certain height, we could use the equation:

$$\mathbf{v^2} = 2 \times \Delta\mathbf{h} \times \mathbf{g}$$

1B: Energy (2. Work and Power)

22) **Efficiency** measures a machine's ability to usefully transfer energy.

**useful energy output = efficiency x total energy input**

As power is energy divided by time, we also get the equation:

**useful power output = efficiency x total power input**